Exercise Sheet 9

In Exercises 9.1–9.3, 9.4(iii) and 9.5, $E = \mathbb{R}^2$ has the standard inner product and $\gamma = (2, 1)$. Write

$$R_B = \{\pm e_1, \pm e_2, \pm e_1 \pm e_2\}, \qquad R_C = \{\pm 2e_1, \pm 2e_2, \pm e_1 \pm e_2\}.$$

Notation follows Lecture 9. Exercises 9.1–9.3 compare B_2 with C_2. Exercises 9.4–9.6 reconstruct the chamber and finiteness arguments.

Exercise 9.1 (Simple roots of B_2 and C_2).

(i) Find the positive and simple roots of R_B and R_C for γ.

(ii) Tabulate $\langle \beta, \alpha^\vee \rangle$ for positive $\alpha, \beta \in R_B$. Explain why (R4) holds for all of R_B.

(iii) Write the positive roots in simple-root coordinates and compute both Cartan matrices $a_{ij} = \langle \alpha_j, \alpha_i^\vee \rangle$.

Exercise 9.2 (Coroots and the isomorphism $B_2 \cong C_2$). An *isomorphism* of root systems $R \subset E$ and $R' \subset E'$ is a linear bijection $T\colon E \to E'$ with $T(R) = R'$ and $\langle T\beta, (T\alpha)^\vee \rangle = \langle \beta, \alpha^\vee \rangle$ for all $\alpha, \beta \in R$.

(i) Show that $\{\alpha^\vee : \alpha \in R_B\} = R_C$.

(ii) For an arbitrary root system R, prove that $R^\vee = \{\alpha^\vee : \alpha \in R\}$ is a root system with the same Weyl group.

(iii) Let $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2)$. Prove $T(R_B) = R_C$, $T s_\alpha T^{-1} = s_{T\alpha}$ and $\langle Tx, (T\alpha)^\vee \rangle = \langle x, \alpha^\vee \rangle$. Find the images of the simple roots and fundamental weights of B_2.

Exercise 9.3 (Weyl group, chambers and walls of B_2).

(i) Compute s_α for $\alpha = e_1, e_2, e_1 - e_2, e_1 + e_2$. Prove that W is the group of signed permutations of the coordinates.

(ii) Describe C, its walls and all chambers. Compare their number with $|W|$.

(iii) Find ω_1, ω_2, P, Q and $[P : Q]$.

(iv) Let $\lambda = (-\frac{1}{2}, \frac{3}{2})$. Check $\lambda \in P$ and find its dominant conjugate by simple reflections.

Exercise 9.4 (Walls of the fundamental chamber). Let R be a root system with simple roots Δ. Fix i and a point x with $(x, \alpha_i) = 0$ and $(x, \alpha_j) > 0$ for all $j \neq i$.

(i) Explain why such x exists. Prove that α_i is the only positive root orthogonal to x.

(ii) Prove that x has a neighbourhood U whose regular points all lie in $C \cup s_i C$, and that U meets both chambers.

(iii) Illustrate (ii) in B_2 for $i = 2$ and $x = (1, 0)$.

Exercise 9.5 (The finiteness lemma in coordinates). Use R_C with the simple roots of Exercise 9.1.

(i) Find ω_1, ω_2, P and Q. Describe P_+ in coordinates.

(ii) Follow the proof of finiteness to list all $\mu \in P_+$ with $\mu \leq 2\omega_1$.

(iii) Show that $\{\mu \in P : \mu \leq 2\omega_1\}$ is infinite. Where does the proof use dominance of μ?

(iv) Show that the norm bound alone does not imply $\mu \leq 2\omega_1$.

Exercise 9.6 (The raising algorithm). Let R be a root system, $\lambda \in P_+$ and $\nu \in W\lambda$.

(i) Prove that $\leq$ is a partial order on E.

(ii) Show that if $\langle \nu, \alpha_i^\vee \rangle < 0$, then $s_i\nu - \nu \in Q_+ \setminus \{0\}$.

(iii) Repeatedly replace ν by $s_i\nu$ for some i with negative label $\langle \nu, \alpha_i^\vee \rangle$. Prove that the process stops at λ. Deduce $w\lambda \leq \lambda$ for all $w \in W$. Which cited result is used?

(iv) Run the algorithm in A_2 for $\nu = -\rho$.